

SVD assumes a matrix R is decomposed to:

$$R = U \cdot \Sigma \cdot V^T$$

$U \Rightarrow$ a matrix where each row corresponds to a users' score on latent factors.

e.g.
$$\begin{bmatrix} 0.58 & 0.62 & -0.62 \\ 0.63 & -0.79 & 0.22 \\ 0.49 & 0.38 & 0.78 \\ 0.22 & -0.16 & -0.07 \end{bmatrix}$$

$\Sigma \Rightarrow$ Tells us how important each latent factor is.

e.g.
$$\begin{bmatrix} 4.7 & 0 & 0 \\ 0 & 2.9 & 0 \\ 0 & 0 & 1.2 \end{bmatrix}$$

$V^T \rightarrow$ a matrix where each column corresponds to an item's scores on latent factors

e.g.
$$\begin{bmatrix} 0.84 & 0.77 & 0.80 & 0.15 & 0.12 \\ 0.10 & -0.08 & 0.05 & 0.93 & 0.89 \\ -0.62 & 0.62 & -0.56 & 0.34 & -0.31 \end{bmatrix}$$

$\Rightarrow$ Σ in Σ quantify how much variance each factor explains.

Simple interp: In V^T , Movie 1-3 are rated highly. If they are known to be... we'd interpret this as action axis.

A user with large score on Factor 1 and small on 2 will be recommended Movies 1-3. firs to

Latent factors $\Rightarrow$ abstract hidden dimensions.